

Solar-Design Tools for Green Building

by Bion Howard

Thousands of residential solar buildings, including many that use multiple solar technologies, have been built in the United States since the 1940s. Today, residential solar-energy applications tend to favor PV electric systems, passive-solar or climate-responsive design, and solar water heating. Surprisingly, solar domestic water heating (SDHW) has been eclipsed by the activity in PV. This is unfortunate, since SDHW is generally cost-effective when coupled with effective backup water heating and energy- and water-efficient distribution systems.

With climate changes and rising energy prices, many designers, builders, remodelers, and homeowners now realize that building-integrated solar can be particularly cost-effective in tandem with super-efficiency, providing measurable savings to owners. The bottom line: Solar energy is back in a big way, but the only means of determining which measures are most cost-effective is to undertake computer energy modeling, preferably with a flexible software tool that provides consistent, understandable results. There are several results-oriented solar-design tools that I have found to be useful in providing practical solutions to common design problems.

CRITICAL ASSISTANCE

Using computer-based design tools definitely helps foster better overall cost-effectiveness for building projects. Imagine trying to make hundreds of different calculations of energy savings, installation costs, thermal loads, and weather variations over and over, all



SUN LIGHT AND POWER

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at the same time! No—can't do it without computers!

Solar-design strategies must be optimized in combination with energy efficiency strategies that are also cost-effective. There are often competing options with different performance claims, some measured and some estimated by vendors, which need to be considered. If building energy efficiency and solar energy designs are not optimized together, then investment economics may become skewed. For example, a larger and higher first-cost solar heating system will be required if an energy-inefficient design is selected for a building's envelope, including its insulation, windows, or foundation.

One of the most important breakthroughs in the evolution of sustainable design, whether it is for buildings, building sites, or other infrastructure, has been the improved effectiveness of interdisciplinary project teams. These project green teams require

good input if they are to make sound decisions. Design tool software helps designers pick through potential minefields of competing claims, and the results provide improved information for project teams.

Computer tools also provide data sources for program compliance verification, such as that required in LEED for Homes, NAHB Green, and Energy Star Homes. Without such information, provided in a manner consistent with sound engineering judgment and environmental principles, it would be virtually impossible to tell a green building from one that is merely slathered with "greenwash."

TOOLS EVOLVE

There are four major types of energy and solar-design tool: those that are used for guidance, calculation, and simulation, and those that are Internet based. In some cases, a given

provider may offer more than one type of tool. Users find it convenient when a single tool or provider offers multiple levels of support including simplified design, code-check capabilities, and Internet connectivity for updates and help screens. Combinations of these capabilities are becoming more common as the energy design tool field evolves.

Guidance. Basic organizational tools provide information to the user from a database that has been compiled from previous analytic, measurement, or calculation results. A guidance tool does not usually make specific calculations or computations in real time. Such tools present tabulated access to database information selected through logical input screens that select appropriate building types, weather locations, fundamental design features, and so on. Then, based on user selections, results are provided in an organized fashion to guide users in further exploration of appropriate measures.

Calculation. These tools usually provide worksheets and tables in which users enter inputs derived from project design and specification assumptions. For example, the user may input the description of a desired solar-energy system, such as a PV array, an inverter or power conditioner, battery backup, building electrical demand characteristics, and utility connections. Then the output results—initial system sizing, for example—are provided from arithmetic calculations or correlation curves embedded in the simplified tool. Results may be displayed in tabular and graphic form. These tools are useful for introductory studies in the building project schematic phase. However, their usefulness is limited by the level of detail included by the software designer.

Simulation. The most advanced class of energy design tools consists of simulation-based software packages. These software packages are based on models employing very detailed engineering computations derived from years of technical analyses of heat transfer, computational fluid dynamics, mechanical engineering, radiant-energy balances, and other fundamental

scientific and physical investigations of how buildings and their energy systems interact with each other and with ambient (outdoor) conditions. Today's simulation software is based on modeling traced back to fundamental building physics studies done at the National Institute for Standards and Technology (formerly NBS) in the early 1970s in response to the energy crisis. This software development supported policymaker needs for engineering decision tools for optimizing insulation levels, window performance, thermal mass, mechanical equipment standards, distribution systems, and the integration of solar-energy systems in buildings. Continued development created DOE-1 and DOE-2; BLAST (U.S. Army Corps of Engineers Building Load Analysis and System Thermodynamics); TARP (Thermal Analysis Research Program); and eventually today's Energy-10 and highly evolved DOE-EnergyPlus software.

Internet based. With the advent of network computing and the Web, Internet-based assessment tools have become increasingly available. Some of these tools provide simple client input forms and, using database software resident on a Web server, also provide quick screening results. Tools of this kind resemble guidance tools, but rather than residing on the user's computer, they use the horsepower of the Web to connect the user with an application. Other Internet tools use the Web as the front-end interface to link with full hourly simulation programs that reside on the host's Web server. These are usually subscription based; the user opens a paid account to access the software, and results from analysis runs appear on a customer-dedicated Web site, accessible with a password. This subscription business model is advantageous for both basic and more sophisticated users, since users do not have to install and maintain complex computer code, and the provider is responsible for upgrades, and for the proper functioning of the tools. My take on this delivery model is that it will continue to gain popularity as long as the user fees are reasonable and the

vendors maintain good product quality and reliability.

CREATIVE TOOLS

Let's review some of my favorite tools that incorporate building energy efficiency analysis and solar-energy design assessment in creative and useful ways. These are all simulation tools; they vary in their level of sophistication and ease of use. They are listed below in alphabetical order.

HEED

In mid-2000, Solar-5 was incorporated into Home Energy Efficient Design (HEED) a user-friendly energy-modeling, and now a solar-design, tool. HEED is user-friendly at all levels of expertise. It is simple enough for beginners; in fact some homeowners have used it to make retrofit decisions on their dwellings. Other users include architects, builders, students, energy program managers, weatherization pros, and green building consultants.

HEED's primary analysis component, Solar-5, is a full annual hourly simulation program. It has been validated compared to DOE-2 and BLAST, using the National Renewable Energy Laboratory (NREL) BEST-EST procedure. HEED self-installs on stand-alone non-networked Windows 95/98 and XP operating systems. A Macintosh OS X version is also available. Climate data are available for over 500 locations. HEED has also been translated into Spanish.

HEED employs a graphic interface to draw a floor plan. Users then click and drag various elements, such as windows, to selected locations. Default lists of standard wall and roof constructions are available and may be edited. A simple base case building can be generated and then subjected to further analysis. Advanced users can use detailed design input options to expand on their building model, and can display output graphics similar to those in Solar-5. While Solar-5 already had many basic features to effectively model basic buildings, when updated to HEED numerous improvements were included. HEED expands on So-

lar-5 features, including editable utility and fuel rates, new rate information for oil and propane, editable air pollution data, attic radiant barriers, operable shading, and enhanced thermal mass algorithms. The HEED update has also eliminated minor software bugs that afflicted Solar-5.

This handy software is available at no cost from the UCLA Department of Architecture and Urban Design, either online by free download at www2.aud.ucla.edu/heed/download.html, or by contacting Research Professor Murray Milne at Milne@ucla.edu.

Note: HEED is represented on the UCLA Web site as a beta release, since it is still undergoing development.

Energy-10

Energy-10 Version 1.8 (with PV) is an advanced capability tool that can be used to help integrate whole-building energy efficiency studies, including PV solar-electric power simulations, and sizing of solar domestic hot water. These solar features were added to Energy-10 during the 2004–2006 development cycle, which was led by software experts at NREL. Since its inception, Energy-10 has provided design support information for small commercial and residential projects with less than 10,000 square feet of conditioned floor area. Some researchers have conducted Energy-10 studies on larger buildings as well. However, results from larger building models may have increased uncertainty due to Energy-10's two-zone limitation (only two thermal zones may be input to the building model, regardless of floor area).

The best feature of Energy-10 is its ability to create a baseline building automatically, and then, through a consistent naming process, produce a large set of prototype variations to test various design strategies. If this is carefully done, a useful set of iterative comparisons of different potential strategies on the same building can be created for analysis. A number of other new features, data library enhancements, and bug fixes accompanied the

recent release of Energy-10 Version 1.8. They are too numerous to include in this short article.

With its new capability to analyze building-integrated PV systems, Energy-10 is one step closer to fully realizing its potential. I hope that the development of this tool will continue, since it serves a large niche in building construction—modeling the myriad small commercial structures that historically have tended to fall through the cracks when it comes to energy and solar design. I would also like to see resumption of the unique and excellent workshop series led by the Sustainable Buildings Industry Council (SBIC). These workshops provided integrated training on sustainable low-energy buildings, always including hands-on use of the Energy-10 software in the sessions. The integration of low-energy buildings design process information with increasing user understanding of energy assessment software provides a valuable set of tools for designers.

Several articles have appeared in *Home Energy* discussing Energy-10 and its evolution over the last ten years into a highly regarded small-building design tool. Energy-10 (not version specific) is also cited in the U.S. Green Building Council's LEED-NC 2.2 Reference

Guide section on effective sustainable low-energy building design. The Energy-10 Web site is www.sbicouncil.org. It has been validated compared to DOE-2 and BLAST, using the NREL BESTEST procedure.

PV DesignPro

Previously available as a stand-alone application, PV DesignPro, by Maui Solar Energy Software Corporation, is now part of a design tool suite called Solar Design Studio, delivered via CD-ROM. Compatible with recent versions of Microsoft Windows, including XP, this tool requires considerable expertise in solar-energy systems design and analysis to wring out maximum results, but it can be used by a novice who is willing to wade through the embedded Help functions.

Inputs include default values and database entries that can be edited and customized. Numerous configurations of panels, connections, and wiring strings may be entered, along with performance parameters for AC inverters, power conditioning, battery charging, and grid interfaces. Numerous climate files are included, along with a weather file generator, to customize for site microclimates.



Computer-based design tools makes it easier to test out various solar-design strategies.

BRUCE WILCOX



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Software design tools are particularly useful for designers who want to incorporate solar and energy efficiency strategies into a project early in the schematic phase.

Available results include, for example, solar-fraction charts, annual- and monthly-performance details, battery charge status, economics estimates based on energy sold, life cycle system costs, and rates of return over the life of the project based on user-editable prices/kWh.

Additional features of the Solar Design Studio include a solar hot water design tool, a climate generator tool, a global insolation database, a video tutorial on CD-ROM, and graphing tools for 3-D plotting of results. Special-order variants include larger-scale PV systems, water pumping, and time-of-use rates, along with a Spanish language version.

The Maui Solar Energy Software Web site is www.maui-solarsoftware.com.

REM/Design Version 12.2

We at Building Environmental Science and Technology have been using the Architectural Energy Corporation (AEC) REM series programs for efficiency policy research, building design assessment, and Home Energy Rating System (HERS) ratings since they were first released in approximately 1989. Now in its twelfth generation,

the Energy and Environmental Builders Association (EEBA) Excellence 2000 Award-winning REM software provides a really useful tool for residential designers, remodelers, project managers, weatherization program managers, and energy professionals. REM/Design calculates heating, cooling, hot water, lights, and appliance loads, fuel and electrical consumption, and costs. System loads information is available by selecting appropriate reports in the output menus. Climate files are included for about 250 North American cities. REM/Design is compatible with Microsoft Windows PC platforms.

The tool uses a specially configured seasonal adaptation of full-annual simulation methods, based on work done by NREL. This reduces runtimes on PC platforms while maintaining good levels of accuracy. Users can provide specific inputs or use a simplified initial approach to start developing a building model. They can then move all their previous work into a detailed model, as more information becomes available. This is particularly useful to designers who want to incorporate energy efficiency into the project early in the

schematic phase, and then improve the details as the project evolves.

The REM/Design tool also provides automatic (and editable) sizing of heating and air conditioning equipment, and it can help to determine whether a project complies with numerous standards and model codes for both the prescriptive and the performance compliance paths. Solar-energy systems can be modeled in conjunction with efficiency improvements, including sunspaces, PV solar-electric systems, and solar water heating. Passive-solar designs are somewhat limited, and users will need to be creative to include high-thermal-mass applications. All the utility rates, component costs, and default building characteristics inputs can be customized by expert users, while the novice or intermediate user can be sure that the default data are very representative. I like REM/Design because it is flexible, powerful, and easy to use for iterative comparisons of different design combinations. The REM/Design Web site is www.archenergy.com/products/rem/. It has been validated compared to DOE-2 and BLAST, using the NREL BEST-EST procedure.

Note: A HERS provider-tuned version of REM (REM/Rate) is also available to certified home energy rating providers. REM/Rate is one of the most widely used HERS tools in the United States. To learn more about it, go to www.archenergy.com/products/rem/rem_rate/.

Another useful HERS tool with similar capabilities is the excellent Florida Solar Energy Center EnergyGauge US software. To learn about this tool, go to www.energygauge.com.

EMERGING SOLAR TOOLS

Here are Web sites where you can learn about two more emerging solar-energy and building efficiency tools: the NREL Solar Benefits model and the MIT Design Advisor. I have not completed detailed reviews of these tools, but they appear promising. They also appear to confirm a trend toward consolidating individual tools into

online design suites that hold to the more scientific and holistic building design paradigm emerging from the sustainable-building revolution.

NREL Solar Benefits

The Solar Benefits Model, developed by NREL, allows solar water heating system designers to vary selected financing, energy production, and cost parameters, to determine the approximate cash flow expected from a particular proposed set of design features.

To learn about the NREL Solar Benefits model, go to www.eren.doe.gov/solarbuildings.

MIT Design Advisor

According to the DOE design tools directory, MIT Design Advisor is a “Web suite of building energy simulators that model energy, comfort, and daylighting performance, and give estimates of the long-term cost of utilities. Energy-load estimates are based on a library of climate data for 30 different cities around the world.”

To learn about the MIT Design Advisor, go to <http://designadvisor.mit.edu>.

Bion D. Howard, of Building Environmental Science & Technology (BEST), provides the buildings industry with technical assistance focused on emerging sustainable designs and technologies. BEST is based in Hilton Head Island, South Carolina (www.energybuilder.com).

For more information:

The DOE Web site for building design software, www.eere.energy.gov/buildings/tools_directory/, currently has over 330 entries. This site provides interesting foraging for PC users seeking well-regarded tools to help with their projects. Each entry has been thoughtfully screened by energy efficiency and green building experts at DOE and their findings are reported in a uniform summary. Although this is one of the best- designed and user-friendly sites at DOE online, sorting through all the possibilities can be a daunting exercise even for a seasoned design tool enthusiast.